

Unified Description of Gravity and Quantum Phenomena through Space Topology: A Rigorous Theory of Vortex Fields

1. Introduction

Modern physics is divided into two fundamental paradigms:
General Relativity (GR) – describes gravity as the curvature of spacetime.
Quantum Field Theory (QFT) – explains particle interactions through the exchange of virtual quanta.
Attempts to unify them (e.g., string theory, loop quantum gravity) face challenges:
Non-renormalizability of gravity at the quantum level.
Lack of clear physical meaning for extra dimensions or discrete spacetime.
The vortex field theory offers an alternative:
Spacetime is a dynamic superfluid medium with vortex excitations.
Particles are stable topological defects (tori, knots, solitons).
Interactions arise from vortex reconfigurations.

2. Mathematical Framework of Vortex Fields

2.1. Space as a Superfluid Condensate

The order parameter of the medium:

$$\psi(\mathbf{r},t)=\sqrt{\rho(\mathbf{r},t)}\,e^{i\theta(\mathbf{r},t)},$$

where:
 ρ is the spatial flow density,
 θ is the phase determining circulation.

Circulation quantization (analogous to charge quantization):

$$\oint \nabla \theta \cdot d\mathbf{l} = 2\pi n, \quad n \in \mathbb{Z}.$$

2.2. Vortex Field Dynamics

The Lagrangian of the system:

$$\mathcal{L} = \frac{\hbar^2}{2m} |\nabla \psi|^2 + V(|\psi|^2) + \frac{1}{2} g^{\mu\nu} (\partial_\mu \psi)(\partial_\nu \psi^*),$$

where $g^{\mu\nu}$ is the metric modified by vortex perturbations.
Euler-Lagrange equation:

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\nabla^2 + V + \lambda|\psi|^2\right)\psi.$$

2.3. Topological Charges and Physical Quantities

Electric charge:

$$q = e \oint \mathbf{v} \cdot d\mathbf{l} = ne, \quad \mathbf{v} = \frac{\hbar}{m} \nabla \theta.$$

Particle mass:

The vortex energy $E = mc^2$ is determined by its stability radius:

$$R = \frac{\hbar}{mc}.$$

3. Gravity in the Vortex Model

3.1. Modified Einstein Equations

The stress-energy tensor includes vortex contributions:

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{(\text{matter})} + T_{\mu\nu}^{(\text{vortex})}\right),$$

where:

$$T_{\mu\nu}^{(\text{vortex})} = \frac{\hbar^2}{m} \left(\partial_\mu \psi \partial_\nu \psi^* - \frac{1}{2} g_{\mu\nu} |\nabla \psi|^2\right).$$

Key difference from GR:

Gravitational "constant" G depends on vortex density:

$$G(N) = G_0 \left(1 + \alpha \ln \frac{N}{N_0}\right).$$

Dark energy arises from vacuum vortex fluctuations.

3.2. Vortex Spacetime Geometry

Metric near a vortex:

$$ds^2 = -\left(1 - \frac{2G(N)M}{c^2 r}\right) dt^2 + \left(1 + \frac{\Phi(r)}{c^2}\right) dr^2 + r^2 d\Omega^2,$$

where $\Phi(r)$ is the vortex field potential.

4. Quantum Phenomena as Vortex Effects

4.1. Uncertainty Principle

Vortex field fluctuations $\delta\theta \sim 1/\delta\rho$ yield:

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}.$$

4.2. Quantum Entanglement

Intertwined vortex threads (analogous to topological QFT):

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|\psi_1\rangle \otimes |\psi_2\rangle + |\psi_2\rangle \otimes |\psi_1\rangle\right).$$

4.3. Tunneling

Vortex mechanism:
A particle crosses barriers via vortex reconfiguration.
Transition probability:

$$P \sim \exp\left(-\frac{2}{\hbar} \int \sqrt{2m(U-E)} \, dx\right).$$

5. Experimental Predictions

5.1. Gravitational Anomalies

Deviations from $1/r^2$ at scales $r < \lambda = \hbar/(mc)$.

High-frequency gravitational waves ($f > 10^3$ Hz) from compact mergers.

5.2. Quantum Systems

Anomalous magnetic moment:

$$\Delta a = \frac{\alpha}{2\pi} + \mathcal{O}(\alpha^2) + \delta a_{\text{(vortex)}}.$$

Superconductivity:

Quantized magnetic flux in vortex lattices.

5.3. Collider Tests

High-energy particle scattering may reveal vacuum vortex structures.

6. Conclusion

The vortex field theory provides:

. **A unified framework** for gravity and quantum phenomena via spacetime topology.

. **Predictions:**

Modified gravity at small scales.

New gravitational wave modes.

. **Applications:**

Controlling vortex states in superconductors.

Topological quantum computing.

. **Critical experiments:**

Submicron-scale gravity measurements.

Detecting vortex excitations in Bose-Einstein condensates.

Further mathematical rigor is required, but this theory opens a path toward a new paradigm in fundamental physics.

References:

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